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SOFTWARE REQUIREMENTS SPECIFICATION
&
SYSTEM ARCHITECTURE DOCUMENT

AI-Powered Inventory & Business Management Platform

Product (Working Title): AI StockFlow / Inventra AI

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1.0	—	Product Team	Initial SRS and architecture outline
2.0	10 Aug 2026	Product Team	Expanded detailed SRS: full functional decomposition, measurable NFRs, interface specifications, risk register, and document control

Review & Approval

Role	Name	Signature	Date
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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the complete functional, non-functional, interface, and architectural requirements for the AI-Powered Inventory & Business Management Platform (the "Platform") to be developed by AI Lead Vision Pvt. Ltd. It serves as the single authoritative reference for the engineering, product management, quality assurance, and delivery teams throughout design, development, testing, deployment, and maintenance. It is also intended to support estimation, sprint planning, and formal acceptance of deliverables.

1.2 Product Scope

The Platform is a commercial, cloud-native, multi-tenant SaaS product that combines a complete inventory, purchase, sales, finance, CRM, HR, and warehouse management suite with a differentiated layer of AI-driven capabilities, including a natural-language business copilot, demand forecasting, automated purchasing, dead-stock detection, price recommendation, computer-vision-based stock operations, a voice assistant, and a composite business health score.

The Platform will initially be deployed for I-ROBOX as the pilot tenant, and will subsequently be licensed to external customers in two market phases: Phase 1 targets small and mid-size retail businesses (toy stores, gift shops, electronics and mobile shops, supermarkets, hardware, clothing, and cosmetic stores); Phase 2 targets warehouses, distributors, importers, manufacturers, pharmaceutical distributors, and FMCG companies. An Enterprise On-Premise Edition is planned for customers with data-residency or regulatory constraints.

1.3 Out of Scope (Current Release)

- Full statutory payroll compliance beyond basic payroll computation (Phase 3+ decision).
- Native integrations with logistics carriers and route execution systems (roadmap item under AI route optimization).
- Marketplace listing management (e-commerce channel sync is a future-phase integration).
- Multi-country tax engines beyond Indian GST at initial launch (architecture must remain extensible).

1.4 Definitions, Acronyms and Abbreviations

Term	Definition
SKU	Stock Keeping Unit — a unique identifier for each distinct product and its variants
SaaS	Software as a Service — centrally hosted subscription software
Tenant	A logically isolated customer organization within the multi-tenant platform
RBAC	Role-Based Access Control
RAG	Retrieval-Augmented Generation — grounding LLM responses in retrieved tenant data
LLM	Large Language Model
PO / GRN	Purchase Order / Goods Receipt Note

Term	Definition
POS	Point of Sale
GST	Goods and Services Tax (India)
MAPE	Mean Absolute Percentage Error — forecast accuracy metric
BFF	Backend-for-Frontend — API layer tailored to each client type
RPO / RTO	Recovery Point Objective / Recovery Time Objective
SLA / SLO	Service Level Agreement / Service Level Objective

1.5 References

- AI Lead Vision Pvt. Ltd. — Product Strategy Note: "AI-Powered Inventory & Business Management Platform" (internal).
- IEEE 830 / ISO/IEC/IEEE 29148 — Guidance for Software Requirements Specifications (structure adapted).
- Applicable Indian GST rules and e-invoicing notifications in force at development time.

1.6 Requirement Prioritization Convention

Each requirement is assigned a priority using the MoSCoW convention: "Must" (mandatory for the stated release), "Should" (important, planned within the release unless constrained), "Could" (desirable, scheduled as capacity allows), and "Future" (explicitly deferred to the roadmap). Unless otherwise stated, requirements labeled Must apply to the MVP (Phase 1) release.

2. Overall Description

2.1 Product Perspective

The Platform is a new, self-contained product. It is not an extension of an existing system, although it will interoperate with third-party systems (payment gateways, messaging providers, accounting packages, and marketplaces) through a documented open API and webhook layer. The Platform follows an API-first, modular architecture: every capability available in the web and mobile clients is exposed through versioned REST APIs, enabling future partner integrations and white-label scenarios.

2.2 Product Functions (Summary)

- End-to-end inventory control: product master, barcodes/QR, serial and batch tracking, multi-warehouse stock, transfers, counts, adjustments, reservations, and bundles.
- Procure-to-pay workflow: purchase requests, purchase orders, goods receipt, purchase returns, and vendor performance scoring.
- Order-to-cash workflow: quotations, sales orders, POS billing, invoicing, returns and credit notes, and delivery documentation.
- Financial visibility: GST-compliant taxation, expense tracking, profit & loss, cash flow, and receivables/payables aging.
- Customer relationship management: leads, follow-ups, customer history, and a sales pipeline.
- Human resources (optional module): employee records, attendance, leave, and optional payroll.
- Warehouse execution: bin management, pick lists, put-away, packing, and dispatch.
- Real-time dashboards and exportable reporting with role-based views.
- An AI layer providing conversational analytics, forecasting, automated purchasing, dead-stock and pricing intelligence, computer vision operations, voice control, and a business health score.

2.3 User Classes and Characteristics

User Class	Characteristics & Typical Usage	Technical Skill
Super Admin (AI Lead Vision)	Manages tenants, subscription plans, billing, feature flags, and platform configuration; monitors platform health	High
Business Owner / Tenant Admin	Full access within the tenant; configures company profile, users, taxes, and approvals; consumes dashboards and AI insights	Low–Medium
Store Manager	Daily inventory, purchasing, and sales oversight; approves transfers and adjustments; acts on AI reorder suggestions	Low–Medium
Sales Staff / Cashier	High-frequency POS billing, sales orders, quotations, and customer lookups; requires a fast, keyboard/scanner-friendly UI	Low
Warehouse Staff	Executes picking, put-away, packing, dispatch, and counts, primarily on mobile devices with barcode scanning	Low

User Class	Characteristics & Typical Usage	Technical Skill
Procurement Officer	Manages suppliers, raises and tracks POs, records goods receipts, handles purchase returns	Medium
Accountant	Manages taxes, expenses, invoices, and financial reports; reconciles receivables and payables	Medium
HR Manager	Maintains employee data, attendance, leave, and optional payroll	Low–Medium
Integration Partner / Developer	Consumes the open REST API and webhooks to integrate external systems	High

2.4 Operating Environment

- Server side: containerized services (Docker) orchestrated on Kubernetes; primary cloud is Microsoft Azure, with AWS and DigitalOcean supported as alternatives; PostgreSQL, Redis, and S3-compatible object storage (MinIO / Azure Blob / AWS S3).
- Web clients: latest two major versions of Chrome, Edge, Firefox, and Safari; responsive layout from 360 px mobile width to large desktop displays.
- Mobile clients: Flutter application targeting Android 10+ and iOS 14+, with camera access for barcode/vision features and microphone access for the voice assistant.
- Peripherals: USB/Bluetooth barcode scanners (keyboard-wedge), thermal receipt printers (ESC/POS), and A4 laser printers for documents.
- Enterprise On-Premise Edition: the same container bundle deployable on customer infrastructure meeting published minimum specifications.

2.5 Design and Implementation Constraints

- Multi-tenancy with strict tenant data isolation is mandatory from the first release; no feature may bypass tenant scoping.
- The technology stack is fixed as: React/Next.js (web), Flutter (mobile), FastAPI/Python (backend), PostgreSQL (database), Redis (cache/queue), S3-compatible object storage, and Python/PyTorch/OpenCV plus LLM and RAG tooling for AI services.
- All external interfaces must be versioned REST APIs secured with token-based authentication; breaking API changes require a new version.
- Indian GST computation, invoice numbering, and document formats must comply with regulations in force at launch.
- AI recommendations that create financial commitments (e.g., automatic purchase orders) must default to human-in-the-loop approval; fully automatic execution is an explicit tenant-level opt-in.
- Competitor price data may be used by the price recommendation engine only where collection and use are legally and contractually permissible.

2.6 Assumptions and Dependencies

- Tenants have reliable internet connectivity; only the POS module is required to tolerate brief offline periods.

- Historical sales data of sufficient depth (ideally 12+ months) materially improves forecasting quality; the system must degrade gracefully for new tenants with sparse data.
- Third-party services (LLM provider, payment gateway, SMS/e-mail/WhatsApp providers) remain available under their published SLAs; the architecture must isolate their failures from core operations.
- The pilot deployment with I-ROBOX will supply real-world feedback prior to general availability.

3. Functional Requirements

Requirements are grouped by module. Each requirement is uniquely identified for traceability into design documents, test cases, and release notes.

3.1 Inventory Management (FR-INV)

Req. ID	Requirement Description	Priority
FR-INV-01	The system shall maintain a Product Master containing SKU code, product name, description, category and sub-category, brand, unit of measure, purchase and selling price, tax class, HSN code, reorder level, images, and configurable custom attributes.	Must
FR-INV-02	The system shall support product variants (e.g., size, color) under a parent product, each with its own SKU and stock position.	Should
FR-INV-03	The system shall generate and print barcodes (EAN-13/Code-128) and QR codes for products, and shall accept scanner input in all stock and billing screens.	Must
FR-INV-04	The system shall optionally track items by serial number (unit-level traceability) and/or batch number with manufacturing and expiry dates, enforced at receipt, transfer, and sale.	Must
FR-INV-05	The system shall maintain stock positions per warehouse and per storage location, with a consolidated tenant-level view.	Must
FR-INV-06	The system shall support stock transfers between warehouses with a configurable approval workflow, in-transit status, and receipt confirmation at the destination.	Must
FR-INV-07	The system shall support cycle counting and full physical audits, generating count sheets, capturing counted quantities (manual or scanned), and producing variance reports with approval-controlled posting of adjustments.	Must
FR-INV-08	The system shall allow manual stock adjustments only with a mandatory reason code, optional supporting document, and a complete audit trail of who adjusted what, when, and why.	Must
FR-INV-09	The system shall reserve stock against confirmed sales orders and exclude reserved quantities from available-to-promise calculations.	Must
FR-INV-10	The system shall support bundle/kit products whose sale automatically deducts component stock according to a defined bill of materials.	Should
FR-INV-11	The system shall raise configurable low-stock alerts per product per warehouse, delivered in-app and via configured notification channels.	Must
FR-INV-12	The system shall maintain a complete, immutable stock ledger (movement history) for every SKU, reconstructable to any point in time.	Must

3.2 Purchase Management (FR-PUR)

Req. ID	Requirement Description	Priority
FR-PUR-01	The system shall support purchase requests raised by staff, routed through a configurable approval chain before conversion to purchase orders.	Should
FR-PUR-02	The system shall create, amend, approve, print/e-mail, and track purchase orders with statuses (draft, approved, sent, partially received, closed, cancelled).	Must
FR-PUR-03	The system shall maintain supplier master data including contacts, GSTIN, payment terms, default lead time per product, and price lists.	Must
FR-PUR-04	The system shall record goods receipts against POs, supporting partial receipts, over/under-receipt tolerances, batch/serial capture, and automatic stock posting.	Must
FR-PUR-05	The system shall process purchase returns (debit notes) against received goods with stock reversal and supplier ledger adjustment.	Must
FR-PUR-06	The system shall compute vendor performance metrics — on-time delivery rate, fill rate, quality rejection rate, and price stability — and display a supplier scorecard.	Should
FR-PUR-07	The system shall maintain a supplier ledger of purchases, payments, and outstanding balances.	Must

3.3 Sales Management (FR-SAL)

Req. ID	Requirement Description	Priority
FR-SAL-01	The system shall support quotations with validity periods, revision history, and one-click conversion to sales orders or invoices.	Must
FR-SAL-02	The system shall manage sales orders with stock reservation, partial fulfillment, and status tracking through delivery and invoicing.	Must
FR-SAL-03	The system shall provide a POS screen optimized for speed: barcode scan-to-add, product search, quantity edit, discounts (line and bill level), multiple payment modes (cash, card, UPI, credit), split payments, and receipt printing.	Must
FR-SAL-04	The POS shall continue billing during brief network outages by queuing transactions locally and synchronizing automatically upon reconnection, with conflict-safe stock posting.	Should
FR-SAL-05	The system shall generate GST-compliant tax invoices with configurable numbering series, and support e-invoice (IRN) generation where applicable to the tenant.	Must
FR-SAL-06	The system shall maintain a customer master with contact details, GSTIN, credit limit, price level, and complete purchase history.	Must
FR-SAL-07	The system shall process sales returns and issue credit notes with stock restoration and customer ledger adjustment.	Must
FR-SAL-08	The system shall generate delivery notes/challans for dispatched goods, linked to the originating order and invoice.	Must

Req. ID	Requirement Description	Priority
FR-SAL-09	The system shall enforce configurable credit limits and overdue blocks on credit customers, with override rights restricted by role.	Should

3.4 Finance (FR-FIN)

Req. ID	Requirement Description	Priority
FR-FIN-01	The system shall compute CGST/SGST/IGST per item based on HSN code, tax class, and place of supply, and produce GST summary reports aligned to return filing (GSTR-1/3B data views).	Must
FR-FIN-02	The system shall record expenses by category with attachments, payment mode, and reporting by period and category.	Must
FR-FIN-03	The system shall generate a Profit & Loss statement for any date range, deriving cost of goods sold from the stock ledger valuation method (weighted average at launch; FIFO configurable later).	Must
FR-FIN-04	The system shall present cash-flow views of inflows (receipts) and outflows (payments and expenses) by period.	Must
FR-FIN-05	The system shall track accounts receivable and payable with aging buckets (0–30, 31–60, 61–90, 90+ days) and automated payment reminders.	Must
FR-FIN-06	The system shall record customer receipts and supplier payments with allocation against specific invoices/bills.	Must
FR-FIN-07	The system shall export finance data in formats consumable by Tally and other accounting packages.	Should

3.5 CRM (FR-CRM)

Req. ID	Requirement Description	Priority
FR-CRM-01	The system shall capture leads with source attribution (walk-in, referral, online, campaign) and assign ownership to users.	Should
FR-CRM-02	The system shall schedule follow-ups with reminders delivered in-app and via configured channels, and log all interactions against the lead/customer timeline.	Should
FR-CRM-03	The system shall present a 360° customer view: profile, orders, invoices, payments, returns, communications, and AI-derived insights (e.g., purchase frequency).	Should
FR-CRM-04	The system shall provide a configurable sales pipeline with stages, values, expected close dates, and conversion reporting.	Could

3.6 Human Resources — Optional Module (FR-HRM)

Req. ID	Requirement Description	Priority
FR-HRM-01	The system shall maintain employee master records including role, department, joining date, documents, and compensation structure.	Could
FR-HRM-02	The system shall record attendance via manual entry, mobile check-in, or integration with biometric devices through the open API.	Could
FR-HRM-03	The system shall manage leave types, balances, requests, and approvals.	Could
FR-HRM-04	The system shall optionally compute payroll from attendance and compensation data, producing payslips; statutory filings remain out of scope for this release.	Future

3.7 Warehouse Operations (FR-WHS)

Req. ID	Requirement Description	Priority
FR-WHS-01	The system shall model warehouse storage as zones, racks, and bins, and record stock at bin level.	Should
FR-WHS-02	The system shall generate pick lists for orders, sequenced by bin location to minimize travel, with scan-confirmed picking on mobile.	Should
FR-WHS-03	The system shall guide put-away of received stock to suggested bins based on product-bin mapping and available capacity.	Should
FR-WHS-04	The system shall manage packing with box/carton records and dispatch with vehicle/courier details, checklists, and gate-pass documents.	Should

3.8 Dashboards & Reporting (FR-RPT)

Req. ID	Requirement Description	Priority
FR-RPT-01	The system shall display a real-time dashboard with today's sales, orders, revenue, gross profit, inventory value, low-stock count, dead-stock value, and top fast-moving items, filterable by store/warehouse and date range.	Must
FR-RPT-02	The system shall render dashboards according to the user's role (e.g., an accountant sees finance widgets; a store manager sees stock and sales widgets).	Must
FR-RPT-03	The system shall provide standard reports for every module (stock summary, stock ledger, sales register, purchase register, GST summary, aging, vendor scorecard) exportable to CSV, Excel, and PDF.	Must
FR-RPT-04	The system shall support scheduled report delivery by e-mail to configured recipients.	Could

4. AI Feature Requirements (Core Differentiator)

The AI layer is the Platform's primary market differentiation. Each AI capability is specified below with its functional behavior and its human-oversight and quality constraints. All AI features operate strictly on the requesting tenant's data; cross-tenant data usage for model responses is prohibited (see Section 8).

4.1 AI Inventory Copilot (FR-AI-COP)

Req. ID	Requirement Description	Priority
FR-AI-COP-01	The system shall provide a conversational interface in which users ask natural-language questions about their business (e.g., "Which products will run out next week?", "Which supplier delivers the fastest?", "Why are sales down?", "Compare this month with last month") and receive answers grounded in the tenant's own data via retrieval-augmented generation.	Must (Ph. 2)
FR-AI-COP-02	Copilot answers shall include the underlying figures and, where applicable, links to the source records or reports used, so users can verify every claim.	Must (Ph. 2)
FR-AI-COP-03	The Copilot shall respect the asking user's RBAC permissions; it shall never reveal data the user could not access through the normal UI.	Must (Ph. 2)
FR-AI-COP-04	The Copilot shall support follow-up questions within a conversation context and allow exporting an answer as a report.	Should (Ph. 2)

4.2 AI Demand Forecasting (FR-AI-FOR)

Req. ID	Requirement Description	Priority
FR-AI-FOR-01	The system shall forecast demand per SKU (and per category) for configurable horizons, including next-month, seasonal, and festival-period demand, using historical sales, trend, and seasonality signals.	Must (Ph. 2)
FR-AI-FOR-02	Forecasts shall carry a confidence indicator, and forecast accuracy (MAPE) shall be tracked and displayed per SKU/category over time.	Should (Ph. 2)
FR-AI-FOR-03	Forecast jobs shall run on a scheduled batch basis with results cached and surfaced in inventory, purchasing, and dashboard views; models shall retrain automatically as new data accumulates.	Must (Ph. 2)
FR-AI-FOR-04	For tenants or SKUs with insufficient history, the system shall fall back to transparent heuristic estimates and clearly label them as such.	Must (Ph. 2)

4.3 AI Auto Purchase (FR-AI-PUR)

Req. ID	Requirement Description	Priority
FR-AI-PUR-01	The system shall recommend purchase quantities per SKU by combining current stock, reserved stock, demand forecast, supplier lead time, and configured safety stock, and shall draft complete purchase orders for review.	Must (Ph. 3)

Req. ID	Requirement Description	Priority
FR-AI-PUR-02	Draft AI purchase orders shall require explicit human approval by default; tenants may opt in, per supplier and value threshold, to fully automatic PO release.	Must (Ph. 3)
FR-AI-PUR-03	Every AI-generated PO shall display its reasoning (inputs and calculation) for auditability.	Should (Ph. 3)

4.4 AI Dead Stock Detection (FR-AI-DSD)

Req. ID	Requirement Description	Priority
FR-AI-DSD-01	The system shall classify inventory as fast-moving, slow-moving, non-moving, or overstocked using configurable velocity thresholds and forecast-aware analysis, and shall quantify the capital locked in each class.	Must (Ph. 2)
FR-AI-DSD-02	The system shall recommend actions for dead/slow stock — e.g., discount levels, bundling, or inter-store transfer — with projected impact.	Should (Ph. 2)

4.5 AI Price Recommendation (FR-AI-PRC)

Req. ID	Requirement Description	Priority
FR-AI-PRC-01	The system shall suggest selling prices per SKU based on target margin, demand elasticity signals, inventory position, and product age; competitor pricing may be included only where its collection and use are legally and technically appropriate for the tenant.	Should (Ph. 3)
FR-AI-PRC-02	Price suggestions shall never be auto-applied; users accept, modify, or reject them, and all applied changes are audit-logged.	Must (Ph. 3)

4.6 AI Image Recognition (FR-AI-VIS)

Req. ID	Requirement Description	Priority
FR-AI-VIS-01	The mobile application shall scan shelves via camera to detect missing/misplaced products and estimate on-shelf counts against the planogram or expected stock.	Should (Ph. 3)
FR-AI-VIS-02	The system shall support camera-based inventory counting during cycle counts, with human confirmation before posting.	Should (Ph. 3)
FR-AI-VIS-03	The system shall provide barcode recognition and OCR from images (e.g., supplier invoices, labels) to accelerate data entry.	Should (Ph. 3)
FR-AI-VIS-04	The system shall flag visibly damaged packages from photos captured at goods receipt.	Could (Ph. 3)

4.7 AI Voice Assistant (FR-AI-VOC)

Req. ID	Requirement Description	Priority
FR-AI-VOC-01	The system shall accept voice commands for common operations — e.g., "Show today's sales", "Add 20 Hot Wheels", "Generate purchase order", "What is my inventory value?" — on web and mobile.	Could (Ph. 3)
FR-AI-VOC-02	Voice-initiated transactions that modify data shall present an on-screen confirmation before committing.	Must (with VOC-01)

4.8 AI Business Health Score (FR-AI-BHS)

Req. ID	Requirement Description	Priority
FR-AI-BHS-01	The system shall compute a composite Business Health Score from inventory health, sales health, cash flow, supplier score, and customer growth, displayed with per-component breakdown, trend, and plain-language improvement suggestions.	Should (Ph. 4)

4.9 Future AI Backlog (Roadmap)

- Chat with ERP (full conversational operation of the system).
- AI-generated narrative reports and AI e-mail drafting.
- AI purchase-negotiation suggestions.
- AI fraud detection and anomaly detection across transactions.
- AI expiry prediction for batch-tracked stock.
- AI warehouse layout optimization and AI delivery route optimization.

5. External Interface Requirements

5.1 User Interfaces

- Responsive web application (React/Next.js) with a consistent design system, dark/light themes, and full keyboard operability of the POS screen.
- Flutter mobile application for Android and iOS covering POS-lite, stock operations (scan, count, transfer, pick/pack), approvals, dashboards, Copilot, and camera/vision features.
- All destructive actions require confirmation; all list views support search, filter, sort, and column export.
- The UI shall be localization-ready (externalized strings); English at launch, additional languages configurable later.

5.2 Hardware Interfaces

- Barcode scanners: USB/Bluetooth keyboard-wedge devices, no driver dependency.
- Receipt printers: ESC/POS thermal printers (network/USB) for POS receipts.
- Document printers: standard OS printing for A4 invoices, POs, and reports.
- Biometric attendance devices: integrated via the open API (push or pull), not by direct driver support.

5.3 Software Interfaces

Interface	Purpose	Direction
Payment gateway (e.g., Razorpay/Stripe)	Subscription billing for the SaaS plans; optional customer payment links	Outbound + webhooks
SMS / E-mail / WhatsApp providers	Alerts (low stock, PO approvals, payment reminders) and OTPs	Outbound
Accounting packages (Tally, QuickBooks)	Export/synchronize vouchers, ledgers, and invoices	Outbound (Phase 2+)
E-commerce marketplaces	Catalog and stock/channel sync for online sellers	Bi-directional (Future)
LLM provider / self-hosted model	Copilot and generative features via an internal AI gateway	Outbound (server-side only)
GST e-invoice (IRP) systems	IRN generation for applicable tenants	Outbound

5.4 Communication Interfaces

- All client–server communication over HTTPS (TLS 1.2+); WebSocket (WSS) for real-time dashboard and notification updates.
- Public REST API, versioned under `/api/v{n}/`, JSON payloads, OAuth2/JWT bearer authentication, and per-tenant rate limits.
- Outbound webhooks with HMAC signatures and retry-with-backoff delivery for key events (order created, stock below threshold, PO approved, invoice paid).

6. Non-Functional Requirements

Non-functional requirements are stated with measurable targets wherever practical, to be validated through load testing, security testing, and operational monitoring.

ID	Category	Requirement / Target
NFR-01	Multi-tenancy	Complete logical data isolation per tenant. Automated tests shall verify that no API, report, export, or AI response can return another tenant's data.
NFR-02	Performance	95th-percentile API response time ≤ 2 seconds for standard CRUD operations and ≤ 1 second for POS item-add under normal load.
NFR-03	Scalability	Horizontally scalable stateless API tier; the architecture shall support growth from one pilot tenant to thousands of tenants without redesign.
NFR-04	Availability	$\geq 99.5\%$ monthly uptime for the production SaaS environment, excluding announced maintenance windows.
NFR-05	Reliability	No committed transaction may be lost; stock and financial postings are atomic and idempotent under retries.
NFR-06	Backup & DR	Automated encrypted backups at least daily; RPO ≤ 24 hours and RTO ≤ 8 hours at launch, tightening with maturity; restores tested quarterly.
NFR-07	Security	TLS 1.2+ in transit; AES-256 (or provider-equivalent) at rest; RBAC on every endpoint; secrets in a managed vault; OWASP Top 10 mitigations verified before release.
NFR-08	Auditability	Immutable audit trail for all stock-affecting, finance-affecting, permission, and configuration changes, retained for the life of the tenant plus the statutory period.
NFR-09	Compliance	Indian GST computation, invoice formats, and e-invoicing at launch; architecture extensible to other jurisdictions.
NFR-10	Usability	A new cashier shall complete a standard POS sale unaided within 10 minutes of first use; core flows require no training manual.
NFR-11	Localization	Multi-currency-ready data model; externalized UI strings; date/number formats per tenant locale.
NFR-12	Offline tolerance	POS continues billing through outages up to 4 hours with local queuing and automatic, conflict-safe synchronization.
NFR-13	Observability	Centralized structured logging, metrics, health checks, distributed tracing, and alerting on SLO breaches.
NFR-14	Maintainability	Modular codebase with $\geq 70\%$ unit-test coverage on core business logic; CI enforces linting, tests, and migration checks.
NFR-15	AI quality	Copilot answers must be grounded in retrieved tenant data with sources displayed; demand-forecast MAPE tracked per tenant with a published improvement target after 6 months of data.

ID	Category	Requirement / Target
NFR-16	AI safety & oversight	AI actions with financial effect default to human approval; all AI recommendations and their acceptance/rejection are logged for evaluation.
NFR-17	Data portability	Tenants can export their complete data (CSV/Excel) at any time; deletion requests honored per contract and law.
NFR-18	Capacity	Support at least 100 concurrent active users per mid-size tenant and 50,000 SKUs per tenant without degradation beyond stated targets.

7. System Architecture

7.1 Architectural Overview

The Platform follows a cloud-native, API-first, modular architecture organized into six layers: (1) Client Layer — web (React/Next.js), mobile (Flutter), and voice interfaces; (2) API Gateway / BFF — authentication, rate limiting, and tenant routing; (3) Core Business Services — FastAPI services for Inventory, Purchase, Sales/POS, Finance, CRM, HR, and Warehouse; (4) AI Engine Layer — forecasting, Copilot (LLM + RAG), computer vision, pricing, and anomaly detection services behind an internal AI Gateway; (5) Integration Layer — payment, messaging, accounting, and marketplace connectors plus outbound webhooks; and (6) Data & Storage Layer — PostgreSQL (tenant-isolated), Redis (cache/queues), S3-compatible object storage, and a vector store for RAG. Cross-cutting Analytics/Reporting and Observability services span all layers.

Client Layer (Web · Mobile · Voice) → API Gateway / BFF (Auth · Rate Limit · Tenant Routing) → { Core Business Services | AI Engine Layer | Integration Layer } → Data & Storage Layer (PostgreSQL · Redis · Object Storage · Vector Store) — with Analytics/Reporting and Observability as cross-cutting layers.

7.2 Architectural Style and Rationale

- Modular monolith at launch: a single deployable FastAPI application with strict internal module boundaries. This maximizes early delivery speed and operational simplicity while preserving a clean extraction path to microservices (AI Engine and POS are the first extraction candidates as load grows).
- API-first: every module capability is exposed through versioned REST endpoints consumed identically by first-party clients and external integrators; webhooks and (later) GraphQL extend the surface.
- Event-driven backbone: domain events (stock changed, order confirmed, PO received) are published to a queue (Redis Streams initially; Kafka when scale requires) to decouple side effects such as alerting, analytics, and AI triggers (e.g., stock-below-threshold → AI reorder suggestion).
- CQRS-lite for reporting: heavy dashboard and report queries read from optimized read models/materialized views rather than transactional tables.

7.3 Multi-Tenancy Model

- Standard tenants: shared PostgreSQL cluster with schema-per-tenant isolation, balancing isolation strength with operational cost.
- Enterprise tenants: optional dedicated database (or dedicated cluster) for customers with stricter isolation, compliance, or performance needs.
- Tenant resolution at the API Gateway via subdomain or JWT claim; the tenant context is mandatory on every internal call and enforced by a shared data-access layer — no query executes without a tenant scope.
- Per-tenant encryption of sensitive columns and per-tenant feature flags for staged AI rollout.

7.4 AI Engine Design

- Internal AI Gateway: a single internal service through which all AI calls flow, providing model routing, prompt management, per-tenant context injection, quota control, response logging, and

provider abstraction (hosted LLM API or self-hosted model can be swapped without touching business services).

- Forecasting service: per-SKU/category time-series models retrained on scheduled batch jobs; outputs persisted to ForecastResult and cached for dashboards and Auto Purchase.
- Copilot service: LLM + RAG over tenant-scoped transactional data and documents held in the vector store; retrieval is filtered by tenant AND by the requesting user's RBAC permissions before generation.
- Vision service: independent, GPU-scalable inference service (PyTorch/OpenCV) for shelf analysis, counting, OCR, and barcode recognition, invoked from mobile camera capture through the AI Gateway.
- Pricing and anomaly services: rule-assisted ML combining margin targets, demand signals, and statistical variance detection.
- Model lifecycle: versioned models, offline evaluation before promotion, per-tenant A/B capability, and continuous accuracy telemetry (e.g., forecast MAPE) feeding retraining decisions.

7.5 Deployment Architecture

- Cloud-native SaaS on Microsoft Azure (primary, aligning with existing Microsoft-stack expertise), with AWS and DigitalOcean as supported alternatives; all services containerized (Docker) and orchestrated on Kubernetes with autoscaling.
- Environments: Development → Staging → Production, promoted through a CI/CD pipeline enforcing automated tests, database migration checks, and security scans.
- Tenant-level feature flags enable gradual rollout of AI features and safe canary releases.
- Enterprise On-Premise Edition: the same container bundle packaged with an installer, offline license activation, and a documented minimum-specification profile; AI features that require external LLM APIs are configurable or substitutable with self-hosted models.

7.6 Technology Stack

Layer	Technology
Frontend (Web)	React, Next.js
Mobile	Flutter (Android 10+, iOS 14+)
Backend / API	FastAPI (Python), Pydantic, SQLAlchemy
Database	PostgreSQL (schema-per-tenant)
Cache / Queue	Redis (cache, sessions, Redis Streams for events)
Object Storage	MinIO / Azure Blob / AWS S3 (images, documents, model artifacts)
AI / ML	Python, PyTorch, OpenCV, LLM integration via AI Gateway, RAG with vector store
Cloud / Orchestration	Azure (primary), AWS, DigitalOcean; Docker + Kubernetes
CI/CD & Observability	Automated pipeline; centralized logging, metrics (Prometheus/Grafana-class), tracing

8. Data Model and Data Management

8.1 Key Entities

Entity	Key Attributes / Notes
Tenant	id, name, plan, region, status, feature flags, isolation mode
User	id, tenant_id, role(s), permissions, MFA status
Product / Variant	id, tenant_id, SKU, category, brand, UOM, HSN, tax class, prices, reorder level, attributes
StockItem	product_id, warehouse_id, bin_id, batch/serial, on-hand qty, reserved qty, valuation
Warehouse / Zone / Rack / Bin	Location hierarchy for bin-level stock
Supplier / Customer	Master data, GSTIN, terms, ledgers, credit limits
PurchaseRequest / PurchaseOrder / GRN / PurchaseReturn	Procure-to-pay documents with line items and statuses
Quotation / SalesOrder / Invoice / CreditNote / DeliveryNote	Order-to-cash documents with line items and statuses
StockMovement	Immutable ledger: type (receipt, sale, transfer, adjustment, return), qty, reason, references, actor, timestamp
Payment / Receipt / Expense	Financial postings with allocations
Lead / Activity / PipelineStage	CRM records
Employee / Attendance / Leave	HR records (optional module)
ForecastResult	product_id, period, predicted demand, confidence, model version
AIRecommendation	type (reorder, price, dead-stock, transfer), payload, reasoning, status (pending / accepted / rejected), actor
AuditLog	Immutable record of sensitive actions

All tenant-scoped tables carry tenant_id (in addition to schema isolation) and are indexed accordingly. Stock and financial documents are append-oriented: corrections occur through reversing entries, never by silent mutation, preserving a complete audit trail.

8.2 Data Retention and Portability

- Transactional and audit data retained for the tenancy's life plus applicable statutory periods.
- Tenants may export their complete data in open formats (CSV/Excel) at any time from within the application.
- Verified deletion of tenant data upon contract termination, subject to legal retention obligations.

9. Security Requirements

- Authentication: e-mail/password with strong hashing, optional multi-factor authentication, session and refresh-token management, and OAuth2/JWT for API access.
- Authorization: RBAC enforced server-side on every endpoint and field-sensitive report; permission changes are audit-logged.
- Tenant isolation: enforced at the data-access layer and verified by automated cross-tenant leakage tests on every release, explicitly including AI Copilot retrieval and responses.
- Encryption: TLS 1.2+ for all transport; encryption at rest for database, backups, and object storage; sensitive fields (e.g., credentials, tokens) additionally encrypted at column level.
- Secrets management: all credentials and API keys held in a managed vault; none in source code or images.
- API protection: per-tenant and per-user rate limiting, input validation, and standard protections against OWASP Top 10 risks (injection, XSS, CSRF, IDOR, etc.).
- AI-specific controls: prompt-injection resistance in Copilot (retrieved documents treated as data, not instructions), output filtering, and no use of tenant data to train models serving other tenants.
- Operational security: least-privilege infrastructure access, dependency and container vulnerability scanning in CI, and periodic third-party penetration testing before major releases.
- Incident response: documented procedure with severity classification, tenant notification obligations, and post-incident review.

10. Integration Requirements

- Payment gateway integration for SaaS subscription billing (plan purchase, renewal, dunning) and optional customer payment links on invoices.
- SMS, e-mail, and WhatsApp Business API notifications for alerts (low stock, PO approval, payment due) and OTPs, with per-tenant channel configuration.
- Accounting export/synchronization for Tally (priority) and QuickBooks.
- E-commerce marketplace catalog and stock synchronization (future phase).
- Public REST API (versioned) and signed webhooks enabling third-party developers and device integrations (e.g., biometric attendance).

11. Development Roadmap and Release Plan

Phase 1 — Core Platform (MVP)

- Modules: Inventory, Purchase, Sales/POS, core Finance, Dashboards & Reporting.
- Foundation: multi-tenant architecture, RBAC, audit trail, open API v1, notification channels.
- Pilot deployment for I-ROBOX; target segment: small retail (toy, gift, electronics, mobile, supermarket, hardware, clothing, cosmetics stores).

Phase 2 — AI Layer v1

- AI Inventory Copilot (grounded Q&A), AI Demand Forecasting, AI Dead Stock Detection.
- AI-enhanced dashboards; accounting export; CRM core.

Phase 3 — AI Layer v2 and Market Expansion

- AI Auto Purchase, AI Price Recommendation, AI Image Recognition, AI Voice Assistant.
- Full Warehouse and HR modules; Enterprise On-Premise Edition.
- Expansion to distributors, importers, manufacturers, FMCG, and pharmaceutical segments.

Phase 4 — Platform Maturity

- AI Business Health Score; fraud/anomaly detection; expiry prediction.
- Warehouse layout and delivery route optimization; partner ecosystem via open APIs; marketplace integrations.

13. Verification, Validation and Acceptance

- Every functional requirement (FR-*) shall be traceable to at least one automated or documented manual test case; a requirements-traceability matrix is maintained.
- Release gates: unit/integration test pass, cross-tenant isolation tests, performance tests against NFR targets, security scan, and migration verification.

- AI features are additionally validated with offline evaluation datasets (forecast accuracy, Copilot answer groundedness) before tenant exposure, and monitored continuously in production.
- Pilot acceptance with I-ROBOX: agreed UAT scenario checklist covering daily store operations end-to-end, signed off before general availability.

14. Risks and Mitigations

Risk	Impact	Mitigation
Sparse historical data at new tenants weakens forecasts	AI value under-delivered early	Transparent heuristic fallbacks; clear confidence labels; value messaging focused on core ERP until data matures
Cross-tenant data leakage (including via AI)	Severe trust/legal damage	Schema isolation + mandatory tenant scoping in data layer + automated leakage tests incl. Copilot
LLM provider cost or availability changes	Feature cost/regression risk	Provider abstraction behind AI Gateway; self-hosted model option; per-tenant quotas
GST/e-invoicing rule changes	Compliance rework	Isolated tax engine module; monitored regulatory calendar
Scope breadth delays MVP	Time-to-market risk	Strict MoSCoW discipline; CRM/HR/WHs 'Should/Could' items deferred without blocking release
On-premise edition support burden	Engineering distraction	Defer to Phase 3; single container bundle; strict minimum-spec policy

15. Open Items for Stakeholder Decision

- Final production cloud provider (Azure vs AWS vs DigitalOcean) and region(s).
- HR/Payroll scope: excluded from MVP entirely, or minimal employee/attendance included?
- GST launch scope: single-state vs multi-state operations, and e-invoice applicability of the pilot tenant.
- Timing of the Enterprise On-Premise Edition relative to Phase 3 engineering capacity.
- LLM strategy for the Copilot: hosted API vs self-hosted model, and the associated data-processing terms.
- Final product name and trademark clearance (AI StockFlow, Inventra AI, SmartStock AI, StockPilot AI, or other).

— End of Document —

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